Coding & Solution

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	5 Marks

Solution Summary

Field	**Details**
Repository Link / URL	https://github.com/lavakumarkavadana1311/HDI_Predictor
Live Deployed URL	https://hdi-predictor-71k5.onrender.com
Programming Language(s)	Python 3.11
Framework(s) Used	Flask, scikit-learn, pandas, Chart.js
Key Features Implemented	HDI + tier prediction, blank-by-default input form, real-country autofill helper, persistent SQLite history with filter/delete/clear, Home dashboard with EDA charts and global stats
Pending / Incomplete Features	User authentication and multi-user private history (identified as a future scalability item, not required for current scope)
Setup / Run Instructions	cd “5. Project Development Phase”; pip install -r requirements.txt; python train_model.py; python seed_history.py; python app.py

Code Quality Checklist

S.No	**Criteria**	**Status**
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The model's R^2 improved from ~0.90 to 0.999 after a single feature-engineering change: log-transforming GNI per capita to mirror the UNDP's own HDI formula, which uses ln(GNI) for its income sub-index.